

PROJECT PROPOSAL

CareGrid.io

A Unified Digital Platform for Hospital Care, Blood & Organ Coordination

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1. Project Overview

CareGrid.io is a comprehensive, web-based healthcare coordination platform designed to unify five critical hospital operations under a single, secure system: organ donation matching, blood bank management, patient monitoring, pharmacy operations, and billing. The platform aims to reduce response time in emergencies, minimize manual record-keeping errors, and improve transparency between hospital staff, patients, and their families.

Team Name	MedMatrix
Project / Website Name	CareGrid.io
Domain	Healthcare Technology — Hospital Management, Blood Bank & Organ Donation
Core Modules	5 (Organ Matching, Blood Bank, Patient/Doctor Dashboard, Pharmacy, Billing)

2. Organ Donation & Compatibility Matching Engine

Since organ matching is time-sensitive and strict, this module adds significant technical depth to the backend and database architecture, ensuring only medically compatible and urgent matches are surfaced to clinicians.

2.1 Compatibility Scoring Algorithm

Automated Matching: Matches donors and recipients based on Blood Group, Tissue Typing (HLA compatibility), Urgent Needs Index, and Geolocation/Distance.

2.2 Recipient Waiting List Priority

Queue Management: Prioritizes recipients using an urgency score (e.g., MELD/PELD score) combined with total time spent on the waiting list.

2.3 Organ Cold-Ischemia Time Tracker

Live Countdown Timers: Tracks preservation windows for organs in transit (e.g., Heart ~4 hours, Kidney ~24 hours) and raises alerts as thresholds approach.

2.4 Anonymous Living Donor Registry

Secure Donor Portal: Allows living donors to register anonymously and complete eligibility screening checklists online.

3. Smart Blood Bank & Emergency SOS

This module digitizes blood inventory management and introduces an automated emergency alert system to accelerate donor mobilization during critical shortages.

3.1 Real-time Inventory & Expiration Alerts

Component-wise Tracking: Monitors Whole Blood, Red Blood Cells, Platelets, and Plasma separately, with automatic notifications when stock levels drop low or units approach expiration.

3.2 Emergency Broadcast System (SOS)

Automated Alerts: Sends automated SMS / Email blasts to registered donors of a specific blood type when an urgent ICU match is needed.

3.3 Donor Eligibility & Interval Tracker

Safety Enforcement: Automatically blocks donors who donated fewer than 56 days ago, in line with medical donation-interval guidelines.

3.4 Geofenced Donor Search

Map-based UI: Displays nearby eligible donors on an interactive map, positioned relative to the requesting hospital.

4. Daily Patient Update & Doctor Dashboard

A centralized dashboard that keeps clinical staff and patients' families informed, reducing the communication burden on ICU and ward staff.

4.1 EHR Timeline & Vitals Logging

Visualized Vitals: Logs daily temperature, blood pressure, oxygen levels, and medication charts, presented as line graphs for quick trend analysis.

4.2 Attendant / Family Portal

Read-only Access: Gives family members a mobile/web view of daily updates written by doctors or nurses, eliminating repeated calls to the ICU desk.

4.3 Ward & Bed Management

Visual Grid Layout: Displays hospital rooms and beds by status — Occupied, Available, Cleaning, or Reserved — for fast allocation decisions.

5. Pharmacy & Prescription Management (E-Prescriptions)

Digitizes the prescription-to-dispensing workflow, reducing manual errors and improving patient safety through automated interaction checks.

5.1 Digital Prescription Builder

Direct Routing: Allows doctors to generate prescriptions digitally, which are automatically sent to the hospital pharmacy.

5.2 Drug Interaction / Allergy Warning

Safety Alerts: Flags dangerous interactions between a newly prescribed drug and a patient's active prescriptions or listed allergies.

5.3 Automated Stock Deduction

Inventory Sync: Automatically reduces internal pharmacy stock whenever medicine is dispensed to a patient.

6. Billing, Insurance & Discharge Workflow

Streamlines the financial and administrative closure of a patient's hospital stay, from itemized billing to discharge documentation.

6.1 Itemized Billing Generator

Automated Calculation: Computes room charges (per day), pharmacy items, surgery fees, and doctor consultation fees into a single itemized bill.

6.2 Insurance Claim Status Tracker

Claims Management: Tracks claims submitted to insurance providers with clear Approval/Rejection status updates.

6.3 Digital Discharge Summary

One-click Generation: Produces the patient's complete stay report, lab test results, and post-discharge medication schedule as a downloadable PDF.

7. Module Summary

#	Module	Key Capabilities
1	Organ Donation & Compatibility Matching	Compatibility scoring, waiting-list priority, ischemia tracking, donor registry
2	Smart Blood Bank & Emergency SOS	Inventory tracking, SOS broadcast, eligibility tracker, geofenced search
3	Patient Update & Doctor Dashboard	EHR vitals timeline, family portal, ward/bed management
4	Pharmacy & Prescription Management	Digital prescriptions, interaction alerts, automated stock deduction
5	Billing, Insurance & Discharge	Itemized billing, claim tracking, digital discharge summary

8. Conclusion

CareGrid.io brings together five interdependent healthcare functions — organ matching, blood bank coordination, patient monitoring, pharmacy management, and billing — into one integrated platform. By automating time-critical decisions such as organ compatibility scoring and emergency blood donor alerts, alongside day-to-day hospital operations like ward management and billing, the system is designed to reduce administrative overhead, minimize human error, and ultimately improve patient outcomes. This document outlines the intended scope of the Software Lab project and will serve as the foundation for subsequent design and development phases.